MCU

+3.3V
1*10uF + 100nF(for each VDD pin)
C25 100n C26 100n C27 100n C28 100n C29 100n C30 100n C31 100n C32 100n C33 100n C34 100n C42 100n C35 10u
GND

S1
TS18-5-15-SL-160-SMT-TR
100n C39
pulled high internally
+3.3V 10k R8
S2
SLW-913535-2A-SMT
RCC_OSC_IN RCC_OSC_OUT

Separate High Freq Filtering for ADCs
+3.3V +3.3VA VDD
C37 10n C38 1u FB1 120R C36 1u

NRST PDR_ON BOOT0 VREF+ PH0 PH1
PG0 PG1 PG2 PG3 PG4 PG5 PG6 PG7 PG8 PG9 PG10 PG11 PG12 PG13 PG14 PG15
PF0 PF1 PF2 PF3 PF4 PF5 PF6 PF7 PF8 PF9 PF10 PF11 PF12 PF13 PF14 PF15
PE0 PE1 PE2 PE3 PE4 PE5 PE6 PE7 PE8 PE9 PE10 PE11 PE12 PE13 PE14 PE15
VCAP_1 VCAP_2
VBAT VDD VDDA VDDSDMMC VDDUSB
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB9 PB10 PB11 PB12 PB13 PB14 PB15
PC0 PC1 PC2 PC3 PC4 PC5 PC6 PC7 PC8 PC9 PC10 PC11 PC12 PC13 PC14 PC15
PD0 PD1 PD2 PD3 PD4 PD5 PD6 PD7 PD8 PD9 PD10 PD11 PD12 PD13 PD14 PD15

I_VAL
BAROMETER_SCLK BAROMETER_MISO BAROMETER_MOSI
USB_OTG_FS_VBUS USB_OTG_FS_D- USB_OTG_FS_D+ SWDIO SWCLK
MAG_SPI_SDI GPS_SCL GPS_SDA Buzzer IMU_SPI_SCLK
IMU_SPI_SDI IMU_SPI_SDO IMU_SPI_CS MAG_SPI_CS BAR_SPI_CS
MAG_SPI_SCK MAG_SPI_SDO IMU_INT1 IMU_INT2
EM_CH4 EM_CH3 EM_CH2 EM_CH1
NAND_SPI_IO3 NAND_SPI_IO0 NAND_SPI_IO1 NAND_SPI_CLK NAND_SPI_IO2 NAND_SPI_CS

USB_5V R18 4.7k USB_OTG_FS_VBUS R17 10k GND

J1
Conn_01x04_Pin
SWDIO SWCLK +3.3V

X1
Crystal
25MHz
RCC_OSC_OUT RCC_OSC_IN
C41 12p C40 12p

VSS VSSA
U1
STM32F767ZIT6

SUB-CIRCUITS

E-match Firing Circuit

+3.7VBATT
SO2B-XASS-1LFSN J7
D7 MBR1020VL
+BATT
GND
Bidirectional TVS diode
CH4 EM_CH4
CH3 EM_CH3
CH2 EM_CH2
CH1 EM_CH1
IVAL I_VAL
File: e-match.kicad_sch

Power Delivery System

+7.4V +7.4VBATT
SO2B-XASS-1LFSN J5
D4 MBRA340
D8 MBR1020VL
GND
Reverse voltage protection + TVS
VIN +3.3V
SolderJumper_2_Open JP1
CR5 PTVS3V3S1UTR_115
File: PDS.kicad_sch

SENSORS

MAGNETOMETER

Used to correct gyro drift(using earth's magnetic field)
U14
MMC5983MA
+3.3V 100n C18 1u C17 GND
MAG_SPI_SCK MAG_SPI_CS MAG_SPI_SDO
SCL/SPI_SCK VDD SPI_CS SPI_SDO GND
spi or i2c
SDA/SPI_SDI INT VDDIO CAP
MAG_SPI_SDI interrupt output +3.3V GND
C15 10u

BAROMETER

GND BAROMETER_MISO BAROMETER_MOSI BAROMETER_SCLK BAR_SPI_CS
PS SDO SDI/SDA SCLK CSB VDD GND
MS5607-02BA
U13
C45 1u C11 100n
+3.3V GND

IMU

U5
LSM6DSVTR
MODE1: Standard 4-wire SPI
IMU_SPI_SDO GND Prog. interrupt IMU_INT1 +3.3V
SDO/SA0 SDX SCX INT1 VDD_IO GND GND
SDA SCL CS SDO_AUX OCS_AUX INT2 VDD
IMU_SPI_SDI IMU_SPI_SCLK IMU_SPI_CS IMU_INT2 Prog. interrupt +3.3V
C24 100n C23 100n 1u C47
GND

BUZZER

30mA
BZ2 Buzzer
Eff. Res. = 400ohms
BOM Consolidation
R23 806R R22 806R R21 4.7k
CR6
Q6 MMBT3904LT3G
C49 1u
Buzzer GND +3.3V
PWM modulation to control loudness

GPS

+3.3V
RF Design 1.6GHz
I2C
R16 4.7k R19 4.7k
RXD TXD SDA SCL EXTINT SAFEBOOT RESET VIO_SEL
VCC VCC_IO V_BCKP VCC_RF LNA_EN RF_IN TIMEPULSE GND
U12
MAX-M10S
R7 10R L2 27n
C19 10n Low noise DC power for active antenna
C44 47pF DC Block Cap
RF_IN RF trace (microstrip)
RSP-122811-02 J3
SIGNAL GND
IPEX / UFL connector
100n C21 10u C22 GPS_SDA GPS_SCL
Critical length = 19.05 cm / 12 = 1.587cm
Trace widths and lengths need to be matched accordingly
Needs a significant physical separation from other noise producing components
Very susceptible to EMI caused by power electronics or digital communication lines

USB-C

MCU needs an external crystal oscillator of 25MHz
48MHz - is configurable in the MCU
Depends on the protocol
J4
USB_C_Receptacle_USB2.0_16P
VBUS CC1 CC2 D- D- D+ D+ SBU1 SBU2 SHIELD GND SH A1
USB_5V R15 4.7k R14 4.7k
USB_OTG_FS_D- USB_OTG_FS_D+
D6 SD05C-01F D5 SD05C-01F
Differential Pair
GND GND

DATA STORAGE

U17
AS5F31G04SND-08LIN
NAND_SPI_CS NAND_SPI_CLK NAND_SPI_IO0 NAND_SPI_IO1 NAND_SPI_IO2 NAND_SPI_IO3
CS# SCLK SI/SIO0 SO/SIO1 WP#/SIO2 HOLD#/SIO3 VCC VSS
+3.3V 100n C16 C46 1u GND

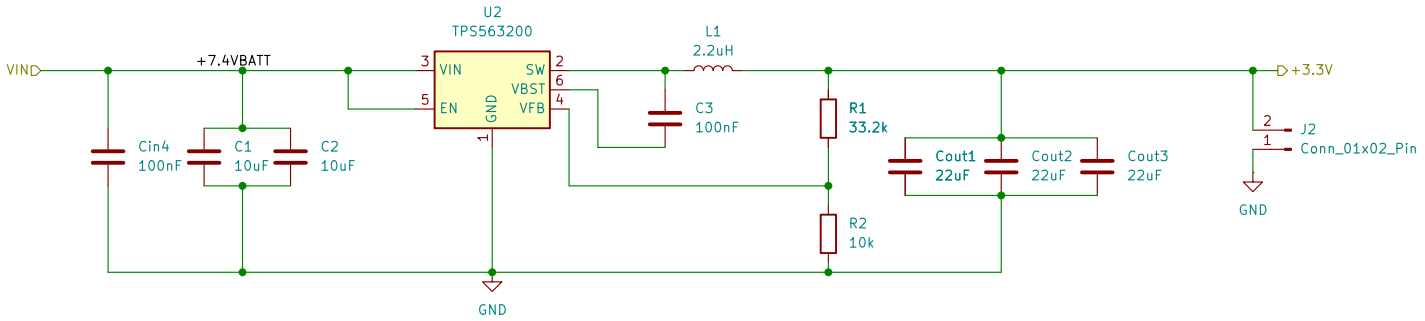
LED indicator

+3.3V R13 806R GND
WL-SMCW_1206
D1
approx 2mA of current

Contributions: Giannis Georgiou, Ian Anderson
Author: Gundeep Sidhu
Sheet: /
File: SRAD_V2.kicad_sch

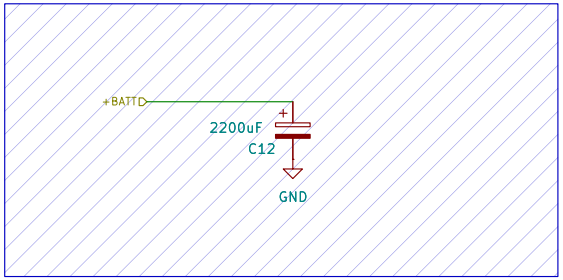
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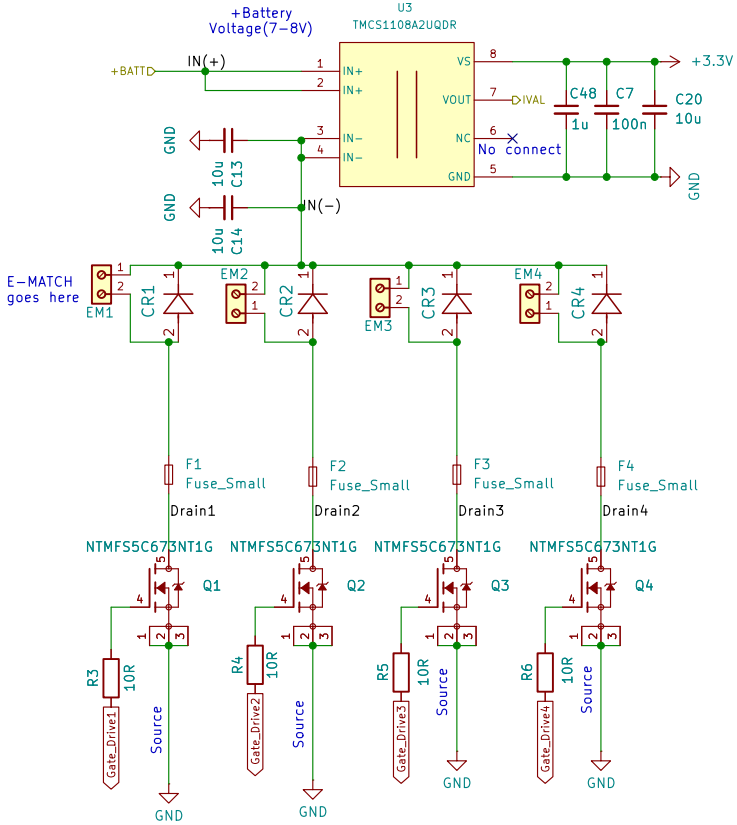
Size: A3	Date: 2026-08-14	**Rev: V2.1**
KiCad E.D.A. 10.0.1		Id: 1/3



Sheet: /Power Delivery System/		
File: PDS.kicad_sch		
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 3/3

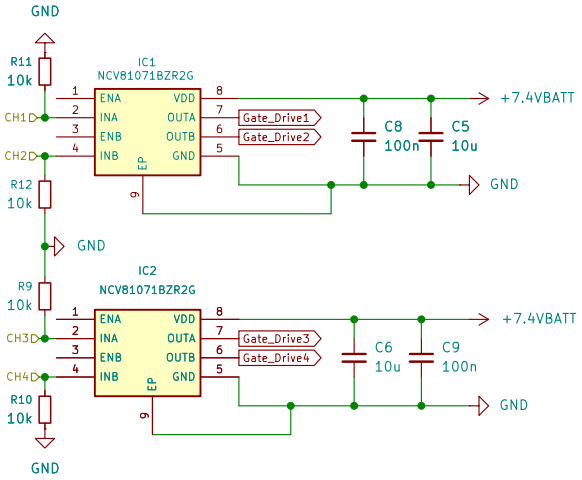
HIGH CURRENT PATH



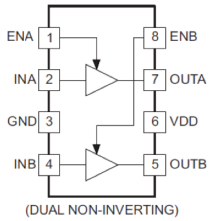


Resistors to limit gate driving current to around 1 Ampere

GATE DRIVERS



Pull down resistors to prevent turning ON during startup



NCV81071B

LED INDICATORS

Sheet: /E-match Firing Circuit/		
File: e-match.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 5/3